

Varen	6869	AgSp1.2	glycine	rm1	1	PGTTPGTITRH	DGKP	KEFI	IP	IGAFTTPGS	FPGPDGR	PIY	VEPAGPGTTPG	AITGPDGKIY		KIIVPTT	QK		70					
Varen	6869	AgSp1.1	glycine	rm1	1								VEPAGPGTTPG	TETGPDGKIY		KVVIP	PTT	PK..GPH	57					
Varen	6869	AgSp1.2	glycine	rm1	2	PGTTPGTITGS	DGKP	TKFI	IP	MGAF	TPGSG	MPGDGR	PIR	VEPAGPGTTPG	VVTRH	DGKIY		KIIVP	AMQMSDE	...	74			
Varen	6869	AgSp1.1	glycine	rm1	2	PGTTPGTVTGR	DGKP	RKFIVP	NGAFTTPGS	FPGPDGE	PI	S	VEPAGPGTTPG	IETDS	DGKI	IK		KVVVP	ST	TPK..GPEGP	75			
Varen	6869	AgSp1.2	glycine	rm1	3	PGTTPGTVTGK														11				
Varen	6869	AgSp1.2	glycine	rm1	10												TIIVPTI			6				
Varen	6869	AgSp1.1	glycine	rm1	11															0				
Varen	6869	AgSp1.2	glycine	rm1	11	PGTTPGTVTGK	DGKP	TKFI	IP	IGAFTTPGS	IPGPDGR	PIR	VEPAGPGTTPG	VVTR	PDGKIY		KIIVPTI	IHMGSDE	...	74				
Varen	6869	AgSp1.2	glycine	rm1	12	PGTTPGTVTGND	DGKP	TKFI	IP	IGAFTTPGS	IPGPDGR	PIR	VEPAGPGTTPG	VVTR	PDGKIY		KIIVPTI	IHMGSDE	...	74				
Varen	6869	AgSp1.2	glycine	rm1	13	PGTTPGTVTGK	DGKP	TKFI	IP	IGAFTTPGS	IPGPDGR	PIR	VEPAGPGTTPG	VVTR	PDGKIY		KIIVPTI	IHMGSDE	...	74				
Varen	6869	AgSp1.2	glycine	rm1	14	PGTTPGTVTGK	DGKP	IKFI	IP	IGAFTTPGS	IPGPDGR	PIR	VEPAG							45				
Mlaby	7048	AgSp1	glycine	rm1b	2	PGTTPGTVTGP	DGKP	TKFIVPT	GAF	TTGSG	SIPGPDGR	PIR	VEPAGPGTTPG	AE	TGPDGKI	N		KIIVVPTT	PKEP		72			
Mlaby	7048	AgSp1	glycine	rm1a	2	PGTTPGTVTGP	DGKP	SKYIVPL	GAV	TTGSG	SIPGPDGR	PIH	VQ	PAGPGTTPG	TI	DSQGN	VD		KIILPTT	PK..GQQ	73			
Mlaby	7048	AgSp1	glycine	rm1a	4	PGTTPG	VVTGPDGQ	P	VQFIVP	QGAFTTPGS	IPGPN	CKRIH	VK	PAGPGTTPG	AK					53				
Mlaby	7048	AgSp1	glycine	rm1a	13	PGTTPG	VVTGPGGEP	L	RFVVPK	GAF	TTGSG	SIPGPN	CKPIH	VG	S	GP	TTTPG	AK	TDS	DGN	TD		76	
Mlaby	7048	AgSp1	glycine	rm2a	21															9				
Mlaby	7048	AgSp1	glycine	rm2a	22	PG	STPGTVTGP	DGMP	P	K										16				
Mlaby	7048	AgSp1	glycine	rm2b	29					GAFTTPGS	IPGPDGR	PIH	VQ	PAG						23				
Ncruc	58532	and13532	AgSp1	glycine	rm	1	PGTTPGTVTGP	DGKP	TKFV	VPL	GAF	TTGSG	SIPGPDGR	PIH	VEPAGPGTTPG	AE	TGPDGKI	VN		KIYLP	PTT	P..KGP	GG	74
Ncruc	58532	and13532	AgSp1	glycine	rm	2	PGTTPGTVTGP	DGKP	TKFV	VPL	GAF	TTGSG	SIPGPDGR	PIH	VEPAGPGTTPG	AE	TGPDGKI	VN		KIYLP	PTT	P..KGP	GG	74
Ncruc	58532	and13532	AgSp1	glycine	rm	3	PGTTPGTVTGP	DGKP	TKFV	VPL	GAF	TTGSG	SIPGPDGR	PIH	VEPAGPGTTPG	AE	TGPDGKI	VN		KIYLP	PTT	P..KGP	GG	74
Ncruc	58532	and13532	AgSp1	glycine	rm	4	PGTTPGTVTGP	DGKP	TKFV	VPL	GAF	TTGSG	SIPGPDGR	PIH	VEPAGPGTTPG	AE	TGPDGKI	VN		KIYLP	PTT	P..KGP	GG	74
Ncruc	58532	and13532	AgSp1	glycine	rm	5	PGTTPGTVTGP	DGKP	TKFV	VPL	GAF	TTGSG	SIPGPDGR	PIH	VEPAGPGTTPG	AE	TGPDGKI	VN		KIYLP	PTT	P..KGP	GG	74
Ncruc	58532	and13532	AgSp1	glycine	rm	6	PGTTPGTVTGP	DGKP	TKFV	VPL	GAF	TTGSG	SIPGPDGR	PIH	VEPAGPGTTPG	VQ	TGPDGK					59		
Mdag	18528	AgSp1.2	glycine	rm	1	PGTTPGTVTGP	DGKP	R												16				
Mdag	18528	AgSp1.2	glycine	rm	16															0				
Mdag	18528	AgSp1.2	glycine	rm	112															0				
Mdag	18528	AgSp1.2	glycine	rm	113	PGTTPGTVTGP	DGKP	KKFV	VPK	GAF	TTGSG	SFSGPDGR	PIH	VEPAGPGTTPG	VQ	TGPDGKI	N		KVVVP	PTT	T..KGP	QGP	75	
Mdag	18528	AgSp1.2	glycine	rm	114	PGTTPGTVTGP	DGKP	KKFV	VPK	GAF	TTGSG	SFSGPDGR	PIH	VEPAGPGTTPG	VQ	TGPDGKI	N		KVVVP	PTT	TK		70	
Mdag	18528	AgSp1.2	glycine	rm	115		SDGKP	QKFV	IPK	GAF	TTTPG	FFPGDGR	PIH	VEPAK						35				
Lcorn	18104	AgSp1.1	glycine	rm1	1	PGTTPGTVTGP	DGKP	IKFIVPT	GAF	TTGSG	SIPGPDGR	PIH	VEPAGPGTTPG	AQ	TGPDGKI	N		KIIVLPTT	..PKAP	QGP	75			
Lcorn	18104	AgSp1.1	glycine	rm1	37	.G	TTTGP	TVTGP	DGKP	TKFIVPK	GAF	TTGSG	SIPGPDGR	PIH	VEPAGPGTTPG	AE	TGPDGKI	N		KIIVVPTT	PK..GP	QGP	74	
Lcorn	18104	AgSp1.1	glycine	rm1	38	PGTTPGTVTGP	DGKP	NKFIVPK	GAF	TTGSG	SIPGPDGR	PIH	VEPAGPGTTPG	AQ	TGPDGKI	N		KIIVLPTT	..PKGP	QGS	75			
Lcorn	18104	AgSp1.1	glycine	rm1	39	PGTTPGTVTGP	DGKP	TKFIVPK	GAF	TTGSG	SIPGPDGR	PIH	VEPAG							45				
Amarm	50219	AgSp1.1	glycine	rm1	1	PGTTPGTIT	TGPDGK	TKFIVPY	GAF	TTGSG	SIPGPDGR	PIH	VEPAGPGTTPG	AE	TGPDGKI	VN		KIYLP	PTT	PK..GP	GGP	75		
Amarm	50219	AgSp1.1	glycine	rm1	2	PGTTPGTVTGP	DGKP	TKFIVPT	GAF	TTGSG	SIPGPDGR	PIH	VEPAGPGTTPG	AE	TGPDGKI	N		KIYLP	ST	..PKGP	GGP	75		
Amarm	50219	AgSp1.1	glycine	rm1	3															0				
Amarm	50219	AgSp1.1	glycine	rm1	8	PGTTPGTVTGP	DGKP	IKFIVPT	GAF	TTGSG	SIPGPDGR	PIH	VEPAGPGTTPG	AE	TGPDGKI	N		KIYLP	ST	..PKGP	GGP	75		
Amarm	50219	AgSp1.1	glycine	rm2	1	PGTTPGTETG	TGDKP	NKFV	VPK	GAF	TTGSG	SIPGPDGR	PIH	VEPAGPGTTPG	AE	TGPDGKI	N		KIIVPTT	TK..APT	GP	75		
Amarm	50219	AgSp1.1	glycine	rm2	78															0				
Amarm	50219	AgSp1.1	glycine	rm2	89															0				
Aarg	AgSp1	FC511	glycine	rm1	1	PGTTPGTIT	TGPDGK	TKFI	IP	LGAFTTPGS	IPGPDGR	PIH	VQ	PAGPGTTPG	AE	TGADGKI	N		KIIVPTT	PKS		71		
Aarg	AgSp1	FC511	glycine	rm1	2	PGTTPGTVTGP	DGKP	KKFV	VPK	GAF	TTGSG	SIPGPDGR	PIH	VEPAGPGTTPG	AQ	TGPDGKI	N		KIIVLPTT	TT	PKGP	VGP	77	
Aarg	AgSp1	FC511	glycine	rm1	3	PGTTPGTVTGS	DGKP	KKFIVPK												22				
Aarg	AgSp1	FC511	glycine	rm1	13					GAFTTPGS	IPGPDGR	PIH	VEPAGPGTTPG	AQ	TGPDGKI	N		KIIVVPTT	TTSKGP	VGP	55			
Aarg	AgSp1	FC511	glycine	rm1	15	PGTTPGTVTGP	DGKP	KKFV	VPK	GAF	TTGSG	SIPGPDGR	PIH	VEPAGPGTTPG	TQ	TGPDGKI	N		K		62			
Aarg	AgSp1	FC511	glycine	rm2	6	PGTTPGTIT	TGPDGNP	TKFIVPL	GAF	TTGSG	SIPGPDGR	PIH	VQ	PAGPGTTPG	AE	TGPDGKI	N		KIILPTT	PKRPSYP	..	75		
Aarg	AgSp1	FC511	glycine	rm2	7	PGTTPGTVTGP	DGQ	P	VKFIVP	QGAFTTPG	TIL	LGPDGR	PIP	VEPQ	G					45				
Atri	AgSp1	genomic	glycine	rm1	1	PGTTPGTIT	TGPDGK	TKFI	IP	LGAFTTPGS	IPGPDGR	PIH	VQ	PAGPGTTPG	AE	TGPDGKI	N		KIIVPTT		68			
Atri	AgSp1	genomic	glycine	rm1	2	PGTTPGTVTGP	DGKP	KKFV	VLPK	GAF	TTGSG	SIPGPDGR	PIH	VQ	PAGPGTTPG	AQ	TGPDGKI	N		KIIVLPTT	TT	PKGP	VGP	77
Atri	AgSp1	genomic	glycine	rm1	47															0				
Atri	AgSp1	genomic	glycine	rm2	40															55				
Mhut	1271	AgSp1	glycine	rm1	1	PGTTPGTIT	TGPDGK	IQFIVPY	GAF	S	TPGS	IPGPDGR	PIH	VEPA	ES	GK	TPG	IETGPDGKIY		KIYLP	PI	IQKGP	DKHRT	77
Mhut	1271	AgSp1	glycine	rm1	2	PGTTPGTIT	TGPDGR	TKFIVPY	GAF	TTPT	CTIPGPDGR	PIH	VEPAGPGTTPG	VERC	D	DGKVY		KIYLP	PLMHIS	P	GELI	76		
Mhut	1271	AgSp1	glycine	rm1	3	PGTTPGTIT	TGPDGK	TKFIVPY	GAF	TTGSG	SIPGPDGR	PIH	VEPAGPGTTPG	VKR	GPDGKVY		KIYLP	PTT	PK..GP	GGP	75			
Mhut	1271	AgSp1	glycine	rm1	4	PGTTPGTITGS	DGRP	IQFIVPY	GAYST	TPGS	IPGPDGR	PIH	VEPAGPGTTPG	IETGPDGKVY		KIYLP	PTT	PK..GP	GGP	75				
Mhut	1271	AgSp1	glycine	rm1	5	PGTTPGTITGS	DGRP	IKFI	IPY	GAYST	TPGS	IPGPDGR	PIH	VEPAGPGTTPG	IETGPDGKVY		KIYLP	PT	RPK	RP	72			
Mhut	1271	AgSp1	glycine	rm1	6	PGTTPGTITGS	DGRP	IKFI	IPY	GAYST	TPGS	IPGPDGR	PIH	VEPAGPGTTPG	VKTGPDGKVY		KIYLP	PTT	PK..GP	GGP	75			
Mhut	1271	AgSp1	glycine	rm1	7	PGTTPGTITGS	DGRP	IKFI	IPY	GAYST	TPGS	IPGPDGR	PIH	VEPAGPGTTPG	VKTGPDGKVY		KIYLP	PTT	PK..GP	GGP	75			
Mhut	1271	AgSp1	glycine	rm1	8	PG	STPGTIT	TGPDGR	TQFI	IP	FGAF	TTGSG	SIPGPDGR	PIH	VEPAG					45				
Mhut	1271	AgSp1	glycine	rm3	17	PGTTPG	VVTDPDGR	Q	VKFIVP	EDAE	TTT	PGVLPGR	NCKPVR	VGPAC						45				

Varen 6869 AgSp1.2 glycine rm1 1	PETHPA GRPIQIV PAG	86
Varen 6869 AgSp1.1 glycine rm1 1	GPGKMFGPMRPRRQRTTPPS VEGPNCKPIQIE PAG	92
Varen 6869 AgSp1.2 glycine rm1 2	QGM...NFATTT IPGPGGRPIQIE PAG	99
Varen 6869 AgSp1.1 glycine rm1 2	GGY...PMRPR.....	83
Varen 6869 AgSp1.2 glycine rm1 3	HMGSD EQGMNFVTTTPIPGPGGRPIQIE PAG	11
Varen 6869 AgSp1.2 glycine rm1 10	GPEG.....PGGYPMGPGNPFPRPMGPGRQRTTPSS VRGPKCKPIQIE PAG	37
Varen 6869 AgSp1.1 glycine rm1 11	QGM...NFATTT IPGPGGRPIQIE PAG	45
Varen 6869 AgSp1.2 glycine rm1 11	QGM...NFATTT IPGPGGRPIQIE PAG	99
Varen 6869 AgSp1.2 glycine rm1 12	QGM...NFATTT IPGPGGRPIQIE PAG	99
Varen 6869 AgSp1.2 glycine rm1 13	QGI...NFATTT IPGPGGRPIQIE PAG	99
Varen 6869 AgSp1.2 glycine rm1 14		45
Mlaby 7048 AgSp1 glycine rm1b 2	TGPGGQPLNPQ..GPISPFPGGGQSM...NPLGPGGS..GG...QTT TPEPIPGPNCKPLQIV PAG	128
Mlaby 7048 AgSp1 glycine rm1a 2	.F.....Y.....FP.....VT...PK PTPTPVTPGPKPIQII PAG	101
Mlaby 7048 AgSp1 glycine rm1a 4		53
Mlaby 7048 AgSp1 glycine rm1a 13	QFPTGPGGQPLNPT..G...LGPRGQPL.....NPLGPGSPGGQTT TPAPVP GP DKPLQIV PAG	91
Mlaby 7048 AgSp1 glycine rm2a 21	PTGPGGQPLNP.....LGPRGQPL.....NPLGPGSPGGQTT TPAPVP GP DKPLQIV PAG	60
Mlaby 7048 AgSp1 glycine rm2a 22		16
Mlaby 7048 AgSp1 glycine rm2b 29		23
Ncruc 58532and13532 AgSp1 glycine rm 1	PGFNF....GTPAT TTPTIPGPGCKPIQVV PAG	103
Ncruc 58532and13532 AgSp1 glycine rm 2	PGFNF....GTPAT TTPTIPGPGCKPIQVV PAG	103
Ncruc 58532and13532 AgSp1 glycine rm 3	PGFNF....GTPAT TTPTIPGPGCKPIQVV PAG	103
Ncruc 58532and13532 AgSp1 glycine rm 4	PGFNF....GTPAT TTPTIPGPGCKPIQVV PAG	103
Ncruc 58532and13532 AgSp1 glycine rm 5	PGFNF....GTPAT TTPTIPGPGCKPIQVV PAG	103
Ncruc 58532and13532 AgSp1 glycine rm 6		59
Msag 18528 AgSp1.2 glycine rm 1		16
Msag 18528 AgSp1.2 glycine rm 16	GPQGPGGYPLGPGGQ...T...T	16
Msag 18528 AgSp1.2 glycine rm 112	RGPQGPGGYPLGPGSPFGPKGPGGQTT TPASVQ GP DKPIQVE PAG	46
Msag 18528 AgSp1.2 glycine rm 113	GG.....YPLGPGSPFGPKGPGGQTT TPASVQ GP DKPIQVE PAG	115
Msag 18528 AgSp1.2 glycine rm 114		70
Msag 18528 AgSp1.2 glycine rm 115		35
Lcorn 18104 AgSp1.1 glycine rm1 1	GI.....NPYYPMRPGR.....	87
Lcorn 18104 AgSp1.1 glycine rm1 37	GL.....NPYYPMGPEGKPM...YPYGPSPFGPQGPGGQT TKTPVP GP DKPIQIV PAG	126
Lcorn 18104 AgSp1.1 glycine rm1 38	GQ.....PL.....FPFGPQ...GPV...GQT TTTPIPGPDCKPIQIV PAG	110
Lcorn 18104 AgSp1.1 glycine rm1 39		45
Amarm 50219 AgSp1.1 glycine rm1 1	GG.....QTT PTPVPGPRCKPIQIL PAG	98
Amarm 50219 AgSp1.1 glycine rm1 2	GG.....QTT PTPVPGPR	88
Amarm 50219 AgSp1.1 glycine rm1 3		10
Amarm 50219 AgSp1.1 glycine rm1 8	GF.....NFA..TP.....AT TTPTIPGPGCKPIQIV PAG	103
Amarm 50219 AgSp1.1 glycine rm2 1	GG.....FPLGP...AGFPLGPGGQPI...YPYGPSPYGPQGPGGKT TTTPIPGPDGEPLSIV PAG	131
Amarm 50219 AgSp1.1 glycine rm2 78	APTGPGGGFPLRT.....TGPGGQPI...YPYGPSPYGPQGPGGK.....	37
Amarm 50219 AgSp1.1 glycine rm2 89	APTGPGGGFPL.....PGPGGQPI...YPYGPSPYGPQGPGGK.....	35
Aarg AgSp1 FC511 glycine rm1 1	PEEPSGRPM...YPFGPGSPYGP...EQTT TTTPIPGPDCKPLQIE PAG	114
Aarg AgSp1 FC511 glycine rm1 2	GG.....MPL..SPYYPQGPGGQPM...NPFAPGSPYGP...EQTT TTTPIPGPDCKPLQIE PAG	129
Aarg AgSp1 FC511 glycine rm1 3		22
Aarg AgSp1 FC511 glycine rm1 13	DG.....QPMYPS..GP.....Q...GPG.GQT TTTPIPGPDCKPIQIE PAG	91
Aarg AgSp1 FC511 glycine rm1 15		62
Aarg AgSp1 FC511 glycine rm2 6	IPT TTTPIPGP ..G QPIQII PAG	96
Aarg AgSp1 FC511 glycine rm2 7		45
Atri AgSp1 genomic glycine rm1 1	PKTPQGS DGQPM ...YPFGPGSPYGP...EQTT TTTPIPGPDCKPLPIE PAG	114
Atri AgSp1 genomic glycine rm1 2	GG.....MPL..SPYSPQGPGGQPM...YPFGPGSPYGP...EQTT TTTPIP	116
Atri AgSp1 genomic glycine rm1 47		13
Atri AgSp1 genomic glycine rm2 40	KGPLGPGGQPM...YPSGPQGTG.....GQT TTTPIPGPDGNPIQIE PAG	97
Mhut 1271 AgSp1 glycine rm1 1	K.....RPSY..THVKDK PIQIV PAG	96
Mhut 1271 AgSp1 glycine rm1 2	LGK ASHCRPIQII PAG	92
Mhut 1271 AgSp1 glycine rm1 3	GFYFG.....MPQY IPGPGGRPIQIV PAG	99
Mhut 1271 AgSp1 glycine rm1 4	GFYFG.....MPQY IPGPGGRPIQIV PAG	99
Mhut 1271 AgSp1 glycine rm1 5	GFYFG.....MPQY IPGPGCKPIQIE PAG	96
Mhut 1271 AgSp1 glycine rm1 6	GFYFG.....MPQY IPGPGCKPIQIE PAG	99
Mhut 1271 AgSp1 glycine rm1 7	GFYFG.....MPQY IPGPDCKPIQIV PAG	99
Mhut 1271 AgSp1 glycine rm1 8		45
Mhut 1271 AgSp1 glycine rm3 17		45

 non-conserved
 similar
 $\geq 50\%$ conserved